Images in a concave and in a convex mirror 1a

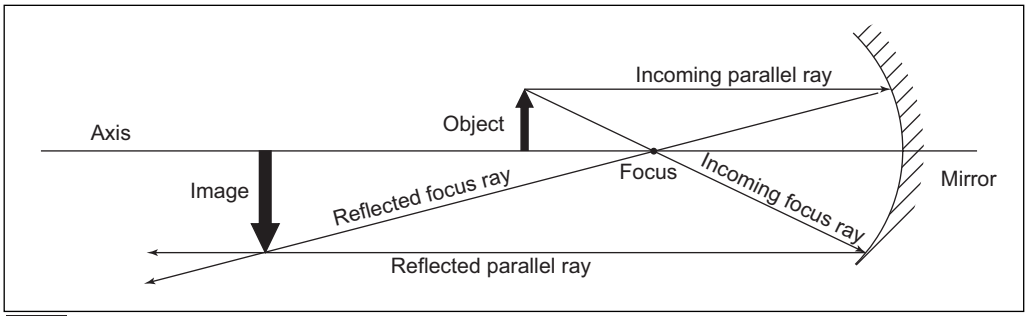
It is possible to predict exactly where the image of an object will appear when that object is in front of a curved mirror. If you select a point on an object (traditionally the tip of a vertical arrow), draw a couple of light rays from that point and follow some rules about reflection, you can find the position of the reflected arrow tip. The position where the two light rays from the object meet or appear to meet after reflection fixes the image.

Here are some things about light rays that you need to know to draw concave mirror diagrams:

- A parallel ray is a ray that is parallel to the axis.
- A focus ray is a ray that travels through the focus.
- An incoming ray travels from the object to the mirror.
- A reflected ray travels away from the mirror.
- After an incoming parallel ray is reflected, it becomes a reflected focus ray.
- After an incoming focus ray is reflected, it becomes a reflected parallel ray.
- If an incoming ray does not hit the mirror, extend it backwards.
- Reflected rays that don't seem to meet might do so if they are extended backwards.
- If reflected rays do not meet, a clear image is not produced.
- If the image is on the same side of the mirror as the object, it is a real image.
- If the image is on the opposite side of the mirror to the object, it is a virtual image.

Compared to its object, an image can be closer to the mirror, further away or the same distance from the mirror. It can be smaller, larger or the same size. It can be upright or inverted, and real or virtual.

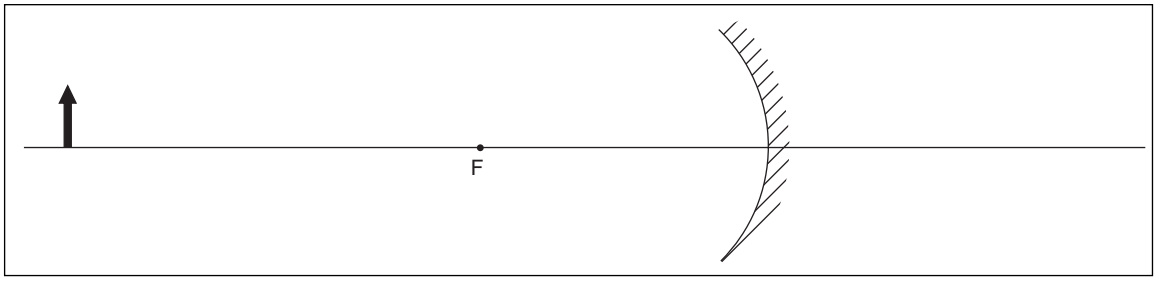
The following diagram shows how drawing a parallel ray and a focus ray from the tip of an arrow can be used to position the reflected tip of the arrow:



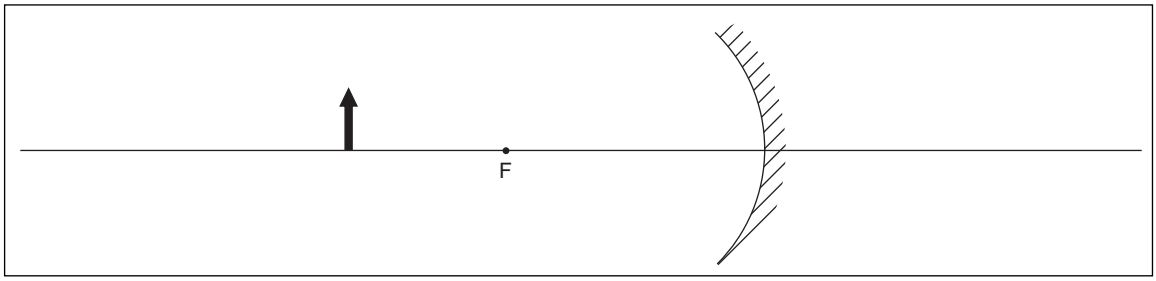
The image is further from the mirror than the object, larger than the object, inverted and real.

1 Construct the image of the object shown in each diagram. Compare image and object.

a



b

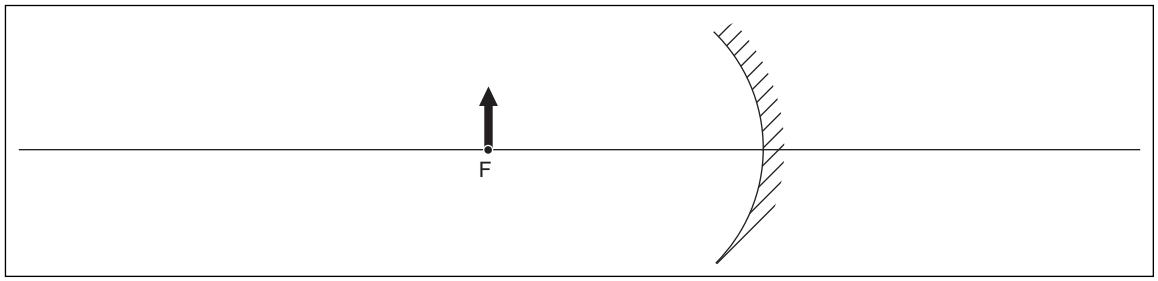


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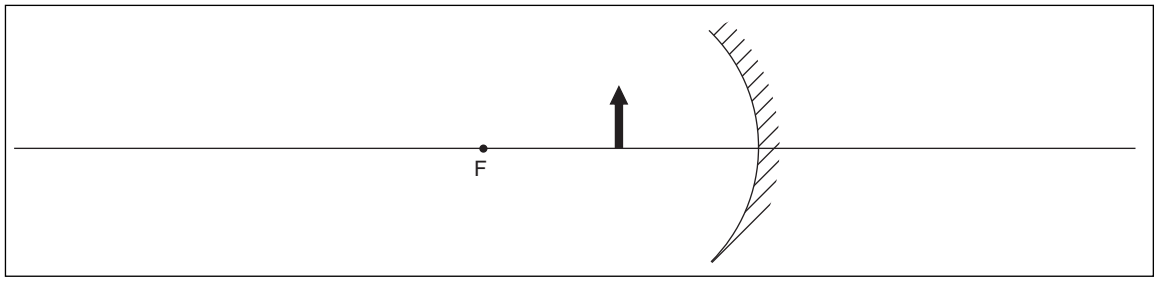
Images in a concave and in a convex mirror

1b

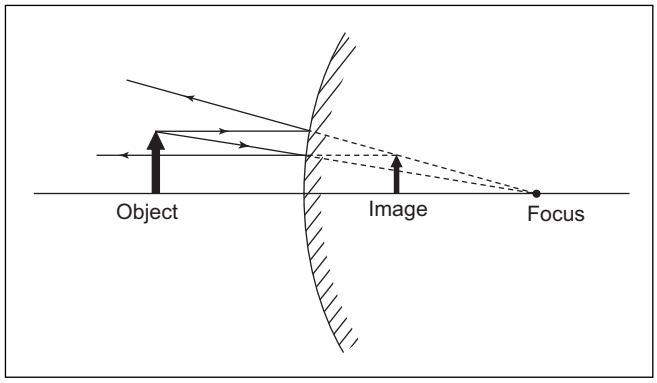
c



d



Below is an object–image diagram for a convex mirror.



The image is closer to the mirror, smaller than the object, upright and virtual.

2 **a** Develop a set of reflection rules to be followed when drawing convex mirrors.

b Draw another object–image diagram for a convex mirror in the box at right. Have the object in a different position to that in the diagram above. Apply your rules. Describe the image.

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